Copyright WILEY-VCH Verlag GmbH & Co. KGaA, 69469 Weinheim, Germany, 2020.

ADVANCED FUNCTIONAL MATERIALS

Supporting Information

for *Adv. Funct. Mater.*, DOI: 10.1002/adfm.202002447

A New Family of Ultralow Loss Reversible Phase-Change Materials for Photonic Integrated Circuits: Sb_2S_3 and Sb_2Se_3

*Matthew Delaney, Ioannis Zeimpekis, Daniel Lawson, Daniel W. Hewak, and Otto L. Muskens**

Copyright WILEY-VCH Verlag GmbH & Co. KGaA, 69469 Weinheim, Germany, 2020.

Supporting Information

Supporting Information to A new family of ultra-low loss reversible phase change materials for photonic integrated circuits: Sb_2S_3 and Sb_2Se3

Matthew Delaney, Ioannis Zeimpekis, Daniel Lawson, Daniel W. Hewak, and Otto L. Muskens *

S1. Temperature dependent ellipsometry

Thin films of Sb_2S_3 and Sb_2Se_3 were crystallized on a hot plate at a range of temperatures for 5 minutes, then ellipsometry used to confirm the phase. **Figure S1a** shows a clear transition between two refractive indices from 260 °C to 280 °C. The film annealed at 270 °C showed partial crystallization under an optical microscope. A longer anneal showed full crystallization at 270 °C, however 5 minutes was picked for consistency across all samples.

Sb_2Se_3 was found to crystallize at a lower temperature of 200 °C (Figure S1b) as compared with Sb_2S_3, with no partial crystallization seen for any sample. In each figure the solid lines represent an amorphous sample, whilst the dashed lines were samples that were visibly crystallized.

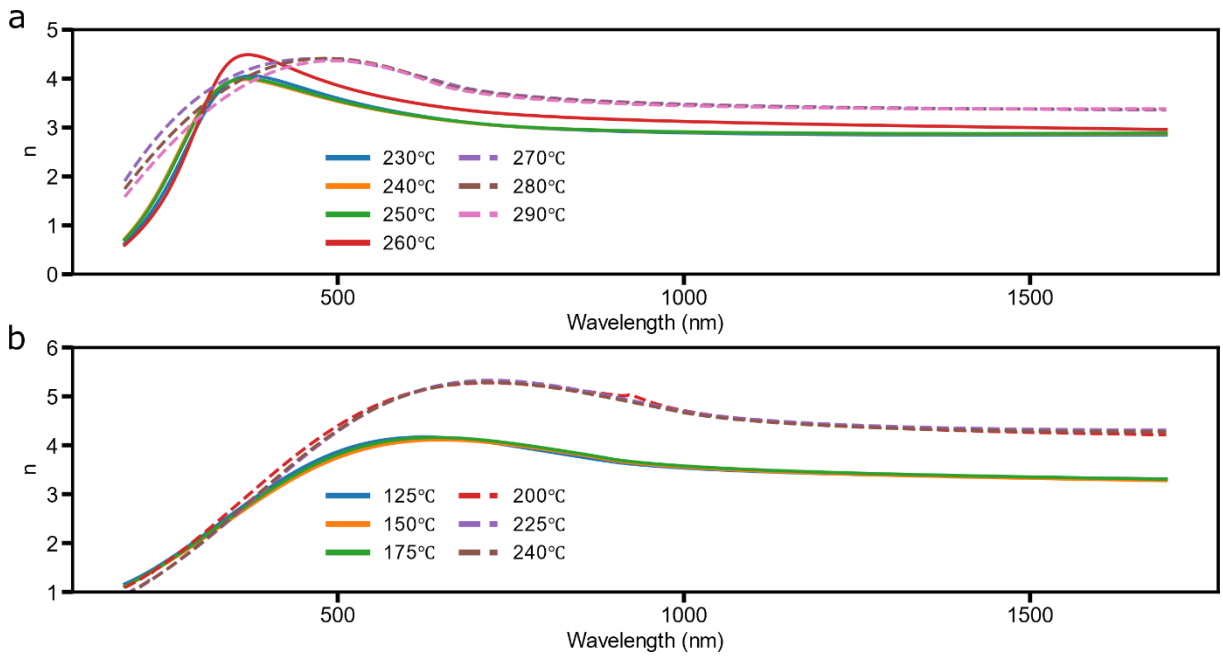

Figure S1 Ellipsometry fits for Sb_2S_3 (a) and Sb_2Se_3 (b) after annealing at different temperatures.

S2. Sb_2S_3 durability test

Figure S2 shows results for a durability test using Sb_2S_3. A decay in reflection change is seen at all points in the durability cycle, with a drop from 5% to 1% seen over the first 2,000 cycles. The reflection change is lower than Sb_2Se_3 due to the lower refractive index. A larger initial change was seen with higher power pulses, suggesting **Figure S2** only shows partial phase change, however the durability of the material with the higher power pulses was reduced to less than 5 cycles before permanent damage occurred.

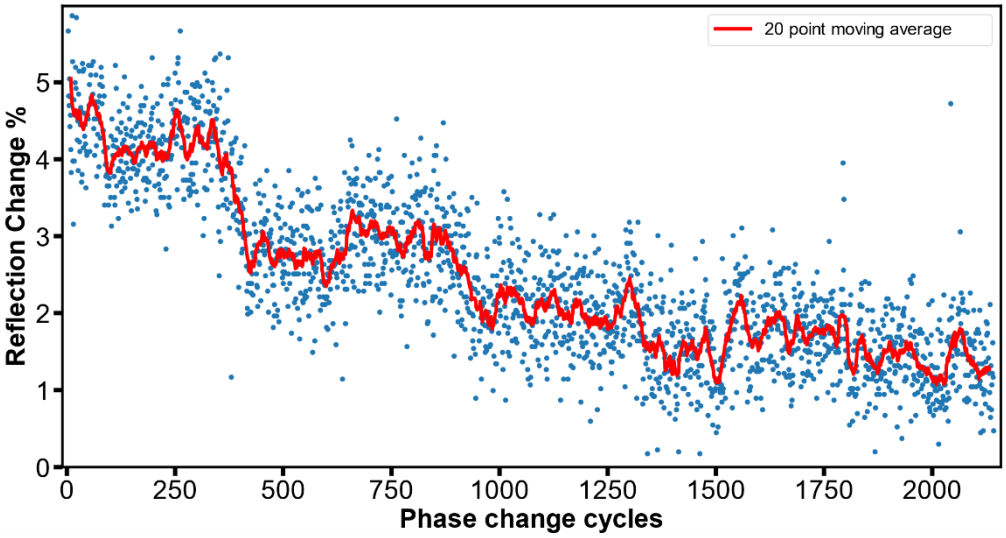

Figure S2 Durability test of Sb_2S_3. The crystallization pulse duration was 2 s at an optical power of 45 mW, while the amorphization pulse duration was 500 ns at an optical power of 80 mW.

S3. Waveguide field overlap results

The mode overlap of the electric fields in the waveguide modes were calculated using the following formula:

$$\eta = \frac{|\int E_1 E_2 dA|^2}{\int |\mathrm{E}_1|^2 dA \int |\mathrm{E}_2|^2 dA}$$

where E_1 and E_2 are the spatially varying electric fields of the two modes corresponding to the 220 nm SOI waveguide with a 25 nm PCM patch (E_1) and without patch (E_1), all waveguides have a SiO_2 capping layer. Table S1 shows the resulting values for the mode overlap η in both amorphous and crystalline phases for the three different PCMs. The values for the crystalline phase correspond to those reported in Figure 7d-f in the main text.

	Cladding Material Phase	
Cladding Material	Amorphous	Crystalline
Sb_2S_3	0.995	0.987
Sb_2Se_3	0.978	0.968
GST	0.956	0.342

Table S1. Values of modal overlap between simulated waveguides with 25 nm claddings of different PCMs compared with the same thickness of silica clad waveguide